
IAL Chemistry Unit 1 — WCH11

Structure, Bonding and Introduction to Organic Chemistry

A* revision and exam-technique guide

Built from analysis of 22 past papers and mark schemes, January 2019 to May 2026

WCH11/01 · 80 marks · 1 hour 30 minutes

Section A: 20 multiple-choice (20 marks) · Section B: 60 marks

40% of the IAS · 20% of the full IAL

1 · What the papers actually look like

The structure was identical in all 22 papers: **Section A is twenty multiple-choice questions worth 20 marks**, and **Section B is 60 marks** spread over four to six structured questions averaging **11.9 marks** each, with the largest running to 24.

Section A is a quarter of the paper. It is the cheapest 20 marks in the qualification and the part most often rushed. There is no penalty for a wrong answer, so never leave one blank.

Timing

80 marks in 90 minutes is just over a minute a mark. A workable split: **22 minutes** on Section A (about a minute per question), **58 minutes** on Section B, **10 minutes** to check. If a multiple-choice question is taking more than 90 seconds, mark your best answer, note the number, and move on.

Command words across all 22 papers

Command	Per paper	What it means here
Calculate	4.6	The dominant command — more calculation-heavy than any other unit in the qualification. Working is credited separately from the answer.
Give / State / Name	9.8 combined	One word or phrase. Formulae, configurations, bond angles, names, observations. Do not explain.
Explain	4.3	Needs a reason. Nearly always ionisation energy trends, bonding, structure or physical properties.
Draw	3.2	Dot-and-cross diagrams, molecular shapes with bond angles, displayed and skeletal formulae, mechanisms with curly arrows, polymer repeat units.
Write	3.1	Equations — balanced, ionic, with state symbols where asked.
Complete	2.0	Tables, electron configurations, partial mechanisms.
Suggest / Deduce	2.2 combined	Unfamiliar by design, or requiring you to work from given data. Never leave blank.

Topic frequency: how many of the 22 papers tested each area

Topic	Papers	Verdict
Moles and amount of substance	22	Every paper. The spine of the unit.
Ideal gas equation, $PV = nRT$	22	
Calculation and data-handling skill	22	
Ionisation energy	21	
Bond angles and molecular shapes	19	Near-certain.
Concentration in mol dm ⁻³	19	
Organic mechanisms	19	
Isotopes	18	
Atom economy	17	Very likely. Note atom economy, not percentage yield — see below.
Polymers and repeat units	17	
Significant figures	17	
Empirical and molecular formulae	16	
Alkenes and addition reactions	14	Common.
Dot-and-cross diagrams	13	
Giant covalent structures	12	
Free radical substitution	10	
Mass spectrometry	10	Periodic.
Structural isomerism	10	
Metallic bonding	9	
Dative covalent bonding	6	
Successive ionisation energies	5	Occasional, but cheap to prepare.
Incomplete combustion and pollutants	5	

Cracking; percentage yield	4 each	See the note below.
Intermolecular forces; electronegativity	1 each	
Titration	0	

THREE FINDINGS WORTH ACTING ON

1. $PV = nRT$ appeared in every single paper. Not most — all 22. If you learn one equation with its units perfectly, make it this one. Section 2.3 deals with it in full.

2. Titration never appeared, and intermolecular forces and electronegativity appeared once each. These belong to other units — titration to the practical paper, intermolecular forces and electronegativity to Unit 2. Revising them for this paper is wasted effort.

3. Atom economy is tested four times as often as percentage yield — 17 papers against 4. Most students prepare the reverse.

2 · Topic 1 — Formulae, equations and amounts of substance

In all 22 papers, and the largest single source of marks. Nothing here is conceptually hard; everything here is lost to unit slips and rounding.

2.1 The core relationships

THE FOUR EQUATIONS

$n = m / M$ moles = mass (g) ÷ molar mass (g mol^{-1})
 $n = c \times V$ moles = concentration (mol dm^{-3}) × volume (dm^3)
 $n = V / 24$ moles of gas at RTP, with V in dm^3 (or V / 24000 with V in cm^3)
 $N = n \times L$ number of particles = moles × Avogadro constant (6.02×10^{23})

Volume conversions cause more lost marks than anything else in the topic: **1 dm³ = 1000 cm³**, so divide cm³ by 1000 before using $n = cV$. The question will usually give you cm³ and want an answer in mol dm⁻³.

2.2 Reacting masses — the standard method

1. Write the **balanced equation**.
2. Convert what you are given into **moles**.
3. Use the **molar ratio** from the equation to find moles of what you want.
4. Convert back into the quantity asked for — mass, volume or concentration.

Where two reactant quantities are given, identify the **limiting reagent** first: divide the moles of each by its coefficient in the equation, and the smaller value is limiting. All subsequent working uses the limiting reagent.

2.3 The ideal gas equation

PV = nRT — IN ALL 22 PAPERS. THE UNITS ARE THE WHOLE DIFFICULTY.

P in pascals (Pa) — if given in kPa, multiply by 1000
V in cubic metres (m³) — if given in cm³ divide by 10⁶; if in dm³ divide by 1000
n in moles · **R** = 8.31 J mol⁻¹ K⁻¹ (given)
T in kelvin — add 273 to a Celsius temperature

Rearrangements you should be able to write without thinking: **$n = PV/RT$, $V = nRT/P$, $T = PV/nR$, $M = mRT/PV$.**

The last of these is how the paper asks you to find a relative molecular mass from gas data, and it has appeared in that form repeatedly. Set out the unit conversions as separate lines before substituting — doing them in your head is where the marks go.

2.4 Empirical and molecular formulae

In 16 of 22 papers.

1. Take the percentage (or mass) of each element.
2. Divide each by its **relative atomic mass**.
3. Divide all results by the **smallest**.
4. Multiply up to whole numbers if needed — ratios ending in .5 double, .33 or .67 treble.
5. For the molecular formula: divide the **relative molecular mass** by the empirical formula mass, then multiply the empirical formula by that whole number.

Water of crystallisation. To find x in a hydrated salt such as MgSO₄·xH₂O: find the mass of water lost on heating, convert to moles, convert the mass of the anhydrous salt to moles, and take the ratio of water to salt. Both this and "find the value of x" phrasing appeared in the papers.

2.5 Percentage yield and atom economy

ATOM ECONOMY — 17 PAPERS, AGAINST 4 FOR PERCENTAGE YIELD

atom economy = (M_r of the desired product ÷ sum of M_r of all products) × 100
equivalently, ÷ the total M_r of the reactants, since mass is conserved.

percentage yield = (actual yield ÷ theoretical yield) × 100, in moles or in mass, provided both are in the same quantity.

The distinction examiners test: **atom economy is a property of the equation** and is fixed before you run the reaction; **percentage yield is a property of the particular run**. A reaction can have 100% atom economy — every addition reaction does, since there is only one product — and still a poor yield.

Why high atom economy matters: less waste produced, so lower disposal costs and less environmental harm; more efficient use of raw materials, which are often finite; and better sustainability. Two of these points is usually the mark.

2.6 Equations and significant figures

- **Ionic equations:** write out the full equation, split all aqueous ionic substances into ions, cancel the **spectator ions** that appear unchanged on both sides, and check that both the atoms and the total charge balance.
- **State symbols** — (s), (l), (g), (aq) — are required whenever the question asks for them, and are a mark in their own right.
- **Significant figures:** 17 of 22 papers specified them. Give the answer to the number asked for, or to the **least precise piece of data** if no instruction is given. Keep full precision through intermediate steps and round only at the end.

3 · Topic 2 — Atomic structure and the periodic table

3.1 Subatomic particles and isotopes

Particle	Relative mass	Relative charge
Proton	1	+1
Neutron	1	0
Electron	1/1836 (negligible)	-1

Isotopes are atoms of the same element with the **same number of protons but different numbers of neutrons**. They have **identical chemical properties**, because chemical behaviour depends on electron configuration, which is the same; their **physical properties differ slightly** because of the mass difference.

RELATIVE ATOMIC MASS FROM ISOTOPIC ABUNDANCES

$$A_r = \Sigma(\text{isotopic mass} \times \text{percentage abundance}) \div 100$$

This appeared in the papers for chlorine, krypton, iron, germanium and bromine. If abundances are given as relative peak heights rather than percentages, divide by the **total** of the heights rather than by 100.

Definition to learn: relative atomic mass is the **weighted mean mass of an atom of an element** relative to **one twelfth of the mass of an atom of carbon-12**.

3.2 Mass spectrometry

In 10 of 22 papers. The stages, in order: **ionisation** (the sample is turned into positive ions), **acceleration** (ions are accelerated by an electric field), **deflection** (by a magnetic field, according to mass-to-charge ratio), and **detection**.

- The **m/z** value of a peak gives the mass of the ion, since the charge is almost always +1.
- For an element, the peaks give the **isotopes** and their **relative abundances**.
- For a molecule, the peak at the highest m/z is the **molecular ion**, and its value is the relative molecular mass.
- **Chlorine** gives peaks in a **3:1** ratio (^{35}Cl and ^{37}Cl) and **bromine** in roughly **1:1**. For Cl_2 the molecular ion region shows peaks at 70, 72 and 74 in a **9:6:1** ratio — a favourite question.

3.3 Electron configuration

Orbitals fill in order of increasing energy: **1s 2s 2p 3s 3p 4s 3d 4p**. Note that **4s fills before 3d but empties first** when the atom is ionised — transition metal ions lose their 4s electrons first, and this is examined.

The two exceptions: chromium is $[\text{Ar}] 3d^5 4s^1$ and copper is $[\text{Ar}] 3d^{10} 4s^1$, because a half-filled or fully filled d sub-shell is more stable.

Orbital capacities: s holds 2 electrons, p holds 6, d holds 10. In box notation, electrons occupy p orbitals **singly first** with parallel spins before pairing, and paired electrons are drawn with **opposite arrows**.

3.4 Ionisation energy

In 21 of 22 papers — the most heavily examined explanation topic in the unit.

DEFINITION — LEARN IT EXACTLY

The **first ionisation energy** is the energy required to remove **one mole of electrons** from **one mole of gaseous atoms** to form **one mole of gaseous 1+ ions**.

Equation form: $X(g) \rightarrow X^+(g) + e^-$

The second ionisation energy removes an electron from the **1+ ion**: $X^+(g) \rightarrow X^{2+}(g) + e^-$. The state symbol (**g**) is compulsory and is frequently the lost mark.

The three factors — every explanation uses these

Factor	Effect
Nuclear charge	More protons means stronger attraction for the outer electron, so a higher ionisation energy
Atomic radius / distance	The further the outer electron is from the nucleus, the weaker the attraction, so a lower ionisation energy
Shielding	More inner shells screen the outer electron from the nuclear charge, so a lower ionisation energy

The trends, and the two dips

- **Down a group** ionisation energy **decreases**: the atomic radius increases and there is more shielding, and these **outweigh** the increase in nuclear charge.
- **Across a period** ionisation energy **increases**: nuclear charge increases while shielding stays essentially the same and the atomic radius decreases.
- **The dip at Group 3** (Al below Mg): the outer electron is in a **p orbital**, which is **higher in energy** and slightly further from the

nucleus than the s orbital it would otherwise occupy, so it is removed more easily.

- **The dip at Group 6** (S below P): the outer p orbital now contains **two paired electrons**, which **repel each other**, so less energy is needed to remove one.

SUCCESSIVE IONISATION ENERGIES AS EVIDENCE

Successive ionisation energies always increase, because each electron is removed from an increasingly positive ion. A **large jump** indicates that the next electron is being removed from a **shell closer to the nucleus** with **less shielding**.

Counting the electrons removed **before** the big jump gives the **number of electrons in the outer shell**, and therefore the **group** the element belongs to. Smaller jumps within a shell reveal the **sub-shells**, which is the evidence for sub-shell structure.

4 · Topic 3 — Bonding and structure

4.1 The three types of bonding

Bond	Definition	Formed between
Ionic	The electrostatic attraction between oppositely charged ions , formed by the transfer of electrons	A metal and a non-metal
Covalent	A shared pair of electrons , attracted to the nuclei of both atoms	Two non-metals
Dative covalent (coordinate)	A covalent bond in which both electrons come from the same atom — one that has a lone pair	NH_4^+ , H_3O^+ , NH_3BF_3 , Al_2Cl_6
Metallic	The electrostatic attraction between positive metal ions and a sea of delocalised electrons	Metals

In a diagram, a dative bond is drawn as an **arrow pointing from the donor atom to the acceptor**. Once formed, it is **identical in strength and length** to any other covalent bond — a point often worth a mark.

4.2 Dot-and-cross diagrams

In 13 of 22 papers. The conventions that carry marks:

- Use **different symbols** for electrons from different atoms, and label which is which.
- For **ionic** compounds, draw **square brackets** round each ion with the **charge** outside, and show the metal ion's outer shell as **empty**.
- Show **only the outer shell** unless told otherwise.
- For **covalent** species, show **all** lone pairs as well as the bonding pairs.
- Check the total electron count against the formula and any charge before you finish.

4.3 Shapes of molecules

Bond angles were asked in 19 of 22 papers, and "lone pair" appeared 115 times across the mark schemes — more than any other content term in the unit.

THE PRINCIPLE, STATED THE WAY THE MARK SCHEME WANTS IT

Electron pairs around the central atom **repel each other** and arrange themselves **as far apart as possible**, to give the minimum repulsion. **Lone pairs repel more strongly than bonding pairs**, so each lone pair **reduces the bond angle by about 2.5°**.

Bonding pairs	Lone pairs	Shape	Bond angle	Example
2	0	Linear	180°	CO_2 , BeCl_2
3	0	Trigonal planar	120°	BF_3 , SO_3
4	0	Tetrahedral	109.5°	CH_4 , NH_4^+
3	1	Trigonal pyramidal	107°	NH_3 , PCl_3
2	2	Bent / V-shaped	104.5°	H_2O
5	0	Trigonal bipyramidal	120° and 90°	PCl_5
6	0	Octahedral	90°	SF_6

When asked to draw a shape, show it in **three dimensions** using wedges and dashed lines, label the **bond angle**, and show any **lone pairs** on the central atom. A flat drawing of a tetrahedral molecule loses the mark.

4.4 Structure and physical properties

Structure	Held together by	Properties, and why
Giant ionic	Strong electrostatic attraction between oppositely charged ions throughout the lattice	High melting temperature, as much energy is needed to overcome the attractions. Conducts when molten or dissolved but not when solid , because the ions are only then free to move. Brittle.
Simple molecular	Strong covalent bonds within molecules, weak intermolecular forces between them	Low melting temperature, because only the weak intermolecular forces are overcome on melting — not the covalent bonds. Does not conduct.
Giant covalent diamond, SiO_2	A continuous network of strong covalent bonds	Very high melting temperature; hard; does not conduct, as there are no free charged particles.
Graphite	Covalent bonds in layers; weak forces between layers	Conducts , because each carbon bonds to only three others, leaving one delocalised electron per atom free to move. Soft and slippery ,

		as the layers slide over one another.
Giant metallic	Attraction between positive ions and delocalised electrons	Conducts in all states, because the delocalised electrons are mobile. Malleable , because the layers of ions slide without breaking the bonding. Melting temperature rises with the charge on the ion and the number of delocalised electrons .

THE SENTENCE THAT LOSES MOST MARKS IN THIS TOPIC

"The covalent bonds break when it melts." For a **simple molecular** substance, melting and boiling overcome the **intermolecular forces only**. The covalent bonds are untouched. Write "intermolecular forces are overcome", and the mark is safe.

5 · Topic 4 — Organic chemistry, alkanes and alkenes

5.1 Nomenclature, formulae and isomerism

- **Naming:** find the **longest carbon chain** containing the functional group; number from the end giving the **lowest** locants; list substituents **alphabetically**; use di-, tri-, tetra- for repeats; separate numbers with commas and numbers from letters with hyphens.
- **Formula types:** *empirical* (simplest ratio), *molecular* (actual numbers), *general* (C_nH_{2n+2} for alkanes, C_nH_{2n} for alkenes), *structural* (condensed, e.g. CH_3CH_2OH), *displayed* (every bond shown), *skeletal* (carbon chain as lines, hydrogens omitted).
- **Structural isomerism:** *chain* (branching differs), *position* (functional group in a different place), *functional group* (different group entirely).

E/Z ISOMERISM

Arises because there is **restricted rotation about the C=C double bond**, and each carbon of the double bond carries **two different groups**.

Assign using **atomic number**: the higher-priority group on each carbon is the one attached through the atom of **higher atomic number**. If the two priority groups are on the **same side** it is **Z**; on **opposite sides**, **E**. *Cis* and *trans* may be used only when each carbon carries one hydrogen.

5.2 Alkanes and free radical substitution

In 10 of 22 papers. The mechanism must be written in three labelled stages, and the conditions stated.

CHLORINATION OF METHANE — CONDITIONS: ULTRAVIOLET LIGHT

Initiation $Cl_2 \rightarrow 2Cl\cdot$ (homolytic fission, caused by UV light)

Propagation $Cl\cdot + CH_4 \rightarrow \cdot CH_3 + HCl$

$\cdot CH_3 + Cl_2 \rightarrow CH_3Cl + Cl\cdot$

Termination any two radicals combining, e.g. $\cdot CH_3 + Cl\cdot \rightarrow CH_3Cl$

Homolytic fission is the breaking of a covalent bond so that **one electron goes to each atom**, forming two radicals. Show the unpaired electron as a **dot** — omitting it loses the mark. Note that propagation **regenerates** a chlorine radical, which is why the reaction is a chain. **Why it is a poor synthetic method:** it gives a **mixture of products**, since further substitution produces CH_2Cl_2 , $CHCl_3$ and CCl_4 , and substitution can occur at different positions in longer chains.

5.3 Crude oil, cracking and combustion

- **Fractional distillation** separates crude oil by **boiling temperature**, which depends on chain length. Longer chains have stronger forces between molecules and condense lower in the column.
- **Cracking** breaks long-chain alkanes into shorter, more useful alkanes and alkenes. *Thermal* cracking uses high temperature and pressure and favours alkenes; *catalytic* cracking uses a **zeolite** catalyst at lower temperature and favours branched alkanes and aromatics for motor fuel.
- **Complete combustion** gives carbon dioxide and water. **Incomplete combustion**, in a limited oxygen supply, gives **carbon monoxide** — toxic, as it binds to haemoglobin in place of oxygen — and **carbon particulates**.
- **Other pollutants:** **sulfur dioxide** from sulfur impurities, causing acid rain; **oxides of nitrogen**, formed from nitrogen and oxygen at the high temperatures inside an engine, causing acid rain and photochemical smog; unburnt hydrocarbons.
- **Catalytic converters** remove these, for example $2CO + 2NO \rightarrow 2CO_2 + N_2$.

5.4 Alkenes and electrophilic addition

Alkenes in 14 papers, mechanisms in 19. The C=C bond consists of a **sigma bond** and a **pi bond**; the pi bond is an area of **high electron density**, which is why alkenes are attacked by **electrophiles** — electron pair acceptors.

DRAWING THE MECHANISM — EVERY FEATURE IS A MARK

1. A curly arrow from the **C=C double bond** to the **$\delta+$ atom** of the attacking molecule.
2. A curly arrow from the **bond being broken** (e.g. H-Br) to the **more electronegative atom**, showing heterolytic fission.
3. The **carbocation intermediate**, with the **positive charge on the correct carbon**.
4. A curly arrow from the **lone pair or negative charge** of the nucleophile to the **positively charged carbon**.
5. The product.

A curly arrow shows the movement of a **pair of electrons** and must start from a **bond, a lone pair or a negative charge** — never from an atom symbol. With a non-polar molecule such as Br_2 , state that the dipole is **induced** as the molecule approaches the electron-rich double bond.

Reagent and conditions	Product	Notes
Br_2 , room temperature	Dibromoalkane	Bromine water is decolourised, orange to colourless — the test for a C=C bond
HBr or HCl	Halogenoalkane	Markovnikov applies — see below
Concentrated H_2SO_4 , then water	Alcohol	Two steps, via an alkyl hydrogensulfate
Steam, phosphoric acid catalyst,	Alcohol	Direct hydration — the industrial route to ethanol

high temperature and pressure		
H ₂ , nickel catalyst, about 150 °C	Alkane	Hydrogenation

MARKOVNIKOV — AND THE REASON, WHICH IS THE A* MARK

When an unsymmetrical alkene reacts with an unsymmetrical reagent, the **major product** is the one formed via the **more stable carbocation**.

Carbocation stability: **tertiary > secondary > primary**, because **alkyl groups are electron-donating** and so **reduce the positive charge on the carbon**, stabilising the ion. The more alkyl groups attached to the positive carbon, the more stable it is. Stating the stability order without that reason scores only half the marks — "carbocation" appeared 70 times across the mark schemes.

5.5 Addition polymerisation

In 17 of 22 papers. Many alkene monomers join by addition to form a long-chain polymer; the **double bond opens** and no other product is formed, so the atom economy is **100%**.

- To draw the **repeat unit**: open the double bond, draw the two carbons joined by a single bond, keep all the substituents in place, and put **extended bonds through the brackets** with **n** outside. Omitting the extended bonds or the *n* loses the mark.
- To find the monomer from a polymer: take one repeat unit and put the double bond back.
- Disposal**: landfill wastes space and most addition polymers are **non-biodegradable**, since the C-C backbone is unreactive; **incineration** releases energy but produces CO₂ and, from PVC, toxic HCl; **recycling** conserves finite crude oil but requires sorting. **Feedstock recycling** breaks polymers back into monomers or other useful chemicals.

6 · Mark-scheme phrase bank

The left column is what scores; the right is the near-miss that does not.

Context	Write this	Not this
First ionisation energy	"Energy to remove one mole of electrons from one mole of gaseous atoms to form one mole of gaseous 1+ ions "	"Energy to remove an electron"
Ionisation energy down a group	" Larger atomic radius and more shielding , which outweigh the increased nuclear charge, so the outer electron is less strongly attracted "	"The atoms get bigger"
Ionisation energy across a period	" Nuclear charge increases while shielding stays similar and the atomic radius decreases "	"There are more protons"
Dip at Group 3	"The outer electron is in a p orbital , which is higher in energy than the s orbital, so it is removed more easily"	"It's a different shell"
Dip at Group 6	"The outer p orbital contains a pair of electrons which repel , so less energy is needed to remove one"	"The electrons are paired"
Large jump in successive IEs	"The next electron is removed from a shell closer to the nucleus with less shielding "	"It is harder to remove"
Molecular shape	"Electron pairs repel and go as far apart as possible ; lone pairs repel more than bonding pairs , reducing the angle by about 2.5° each"	"The lone pair pushes the bonds down"
Ionic bonding	" Electrostatic attraction between oppositely charged ions "	"The ions stick together"
Metallic bonding	" Electrostatic attraction between positive metal ions and delocalised electrons "	"A sea of electrons"
Melting a simple molecular solid	" Intermolecular forces are overcome "	"The covalent bonds break"
Why graphite conducts	"Each carbon bonds to only three others, leaving one delocalised electron per atom free to move"	"It has free electrons"
Why ionic solids don't conduct	"The ions are not free to move in the solid, but are when molten or in solution "	"Solids don't conduct"
Isotopes	"Same number of protons , different number of neutrons ; identical chemical properties because the electron configuration is the same"	"Different forms of an element"
Homolytic fission	"The bond breaks so that one electron goes to each atom , forming two radicals "	"The bond splits in half"
Why free radical substitution is a poor synthesis	"It produces a mixture of products — further substitution and substitution at different positions"	"It doesn't work well"
Carbocation stability	" Alkyl groups are electron-donating , so they reduce "	"Tertiary is more stable"

	the positive charge and stabilise the ion: tertiary > secondary > primary"	
Curly arrows	Show the movement of a pair of electrons , starting from a bond, lone pair or negative charge	Arrows from atom symbols or from nowhere
Test for a C=C bond	"Bromine water is decolourised, from orange to colourless "	"It changes colour"
Atom economy benefit	" Less waste , so lower disposal costs and less environmental harm; more efficient use of finite raw materials"	"It's better for the environment"
Why addition polymers persist	"The C-C backbone is unreactive , so they are non-biodegradable "	"They don't rot"

7 · The mistakes that cost the grade

1. **Unit errors in $PV = nRT$** — kPa not converted to Pa, cm³ or dm³ not converted to m³, Celsius not converted to kelvin. This single equation is in every paper.
2. Forgetting that **1 dm³ = 1000 cm³** when using $n = cV$.
3. Rounding at intermediate steps instead of at the end.
4. Ignoring an explicit instruction about **significant figures** — asked for in 17 papers.
5. Not identifying the **limiting reagent** when two reactant quantities are given.
6. Confusing **atom economy** with **percentage yield**.
7. Omitting **state symbols** when the question asks for them.
8. Leaving spectator ions in an **ionic equation**, or not checking the charge balances.
9. Omitting **(g)** from an ionisation energy equation or definition.
10. Omitting **"one mole"** from the ionisation energy definition.
1. Explaining a group trend with nuclear charge alone, without saying that radius and shielding **outweigh** it.
2. Writing the **4s/3d** configuration of an ion by removing 3d electrons first — the **4s** electrons go first.
3. Forgetting the **chromium and copper** exceptions.
4. Giving a bond angle without the **shape**, or the shape without the **angle**, when both were asked for.
5. Drawing a tetrahedral molecule **flat** instead of in three dimensions with wedges.
6. Forgetting lone pairs in a dot-and-cross diagram of a covalent molecule.
7. Omitting **square brackets and charges** in an ionic dot-and-cross diagram.
8. Saying **covalent bonds break** when a simple molecular substance melts.
9. Saying a metal conducts because of "free ions" rather than **delocalised electrons**.
10. Omitting the **dot** on a radical.
1. Writing only one propagation step, or not showing that a radical is **regenerated**.
2. Drawing curly arrows from atoms rather than from **electron pairs**.
3. Putting the **positive charge on the wrong carbon** in a carbocation.
4. Stating Markovnikov's outcome without the **electron-donating alkyl group** reason.
5. Drawing a polymer repeat unit without **extended bonds through the brackets** or without **n**.

8 · Final checklist

Can you, from memory...	Can you, from memory...
• Write all four mole relationships? • State every unit required by $PV = nRT$, and the rearrangement for M_r? • Work a reacting-mass calculation including a limiting reagent? • Find an empirical formula from percentages, and a molecular formula from it? • Find x in a hydrated salt? • Define and calculate atom economy, and give two reasons it matters? • Write an ionic equation and check the charge balance? • Calculate a relative atomic mass from isotopic abundances? • Name the four stages of mass spectrometry? • Explain the 3:1 and 9:6:1 chlorine patterns? • Write the electron configuration of any element to Kr, including Cr and Cu? • Write the configuration of an ion, removing 4s before 3d?	• State the first ionisation energy definition exactly, with the equation? • Explain the trend down a group and across a period, using all three factors? • Explain both the Group 3 and Group 6 dips? • Deduce a group from successive ionisation energy data? • Define ionic, covalent, dative and metallic bonding? • Draw dot-and-cross diagrams for an ionic and a covalent species? • Give all seven shapes with their bond angles? • Explain four structure types and their properties? • Name an organic compound and draw all its structural isomers? • Assign E and Z using atomic number? • Write all three stages of free radical substitution? • Draw electrophilic addition with every curly arrow and the carbocation? • Explain Markovnikov using carbocation stability? • Draw a repeat unit from a monomer, and a monomer from a repeat unit?

- Give three disposal routes for polymers with a drawback of each?

THE SHORT VERSION

This is the most calculation-heavy paper in the qualification — 4.6 "calculate" commands per paper, and three of the four guaranteed topics are numerical. Most marks lost here are lost to units, rounding and significant figures rather than to chemistry. Learn $PV = nRT$ with its units until it is automatic, since it appeared in all 22 papers. Then learn the ionisation energy explanations and the seven shapes with their angles word-perfect, because those are the explanation marks that recur every session. And do not spend time on titration, electronegativity or intermolecular forces: on the evidence of 22 papers, they are not examined here.